

B.Tech Final Year Major Project Proposals

Timeline: Mid-July 2026 to December 2026

Domain: Agentic AI, Generative AI, and Multi-Modal Systems

Team Size: 4 Members

Prepared For: Rhian & Team

Project Option 1: SentinelOps — Autonomous Multi-Agent DevSecOps & Patching Pipeline

The Problem Statement

Existing Static Application Security Testing (SAST) and vulnerability scanners (like SonarQube or Snyk) excel at flagging security flaws, code smells, and vulnerabilities in repositories. However, they are completely passive: they do not fix the code. Developers must manually interpret logs, write patches, and ensure the fixes do not break existing functionality. This creates a massive bottleneck in the software development lifecycle (SDLC), leading to delayed deployments and unpatched production vulnerabilities.

The Agentic AI Solution

Build an autonomous multi-agent system that hooks into a Git repository, detects security vulnerabilities, reasons through the code context, writes code fixes, runs them in a sandboxed environment to verify they pass tests, and opens a verified Pull Request.

Division of Work (4 Modules for 4 Students)

- **Student 1 (Codebase Ingestion & Context Mapper):** Builds the pipeline to ingest repository codebases, parse Abstract Syntax Trees (AST), and construct a Vector Database/Knowledge Graph so the agents understand project dependencies and context.
- **Student 2 (Vulnerability Analysis & Patching Agent):** Interfaces with scanning APIs, interprets security flaws (e.g., OWASP Top 10), and engineers specialized LLM prompts/agents to generate precise code fixes.
- **Student 3 (Automated Sandbox & Testing Agent):** Builds a secure dockerized environment where the modified code is compiled. This agent dynamically generates new unit tests to verify the patch and runs existing test suites to prevent regressions.
- **Student 4 (Orchestrator & DevSecOps Interface):** Designs the central supervisor agent

(using frameworks like LangGraph or CrewAI) to manage state, handle agent-to-agent communication, and build the frontend dashboard/GitHub App integration.

Suggested Tech Stack

Frameworks: LangGraph, CrewAI, or Autogen

LLM Core: Claude 3.5 Sonnet, GPT-4o via API, or Llama-3

Backend & Sandbox: Python, FastAPI, Docker SDK

Database: Qdrant or Neo4j

Project Option 2: ResQNet — Multi-Modal Multi-Agent Command Center

The Problem Statement

During urban emergencies, flash floods, or high-density public events, emergency management forces rely on fragmented data streams. High-resolution CCTV feeds, chaotic social media updates, and emergency calls exist in silos. Existing emergency systems lack the ability to simultaneously parse multi-modal data in real time, cross-reference conflicting reports, evaluate asset availability, and autonomously coordinate optimized, multi-agency dispatch strategies.

The Agentic AI Solution

Develop a centralized command center powered by a team of collaborative AI agents that ingest multi-modal data (video, text, audio) to monitor public safety, predict crowd anomalies and stampede risks, and orchestrate real-time emergency responses. Expanding on foundational concepts like utilizing CCTV feeds for proactive security monitoring, this architecture scales into a comprehensive multi-agent dispatch system.

Division of Work (4 Modules for 4 Students)

- **Student 1 (Multi-Modal Perception Pipeline):** Develops the Vision-Language Model (VLM) and Audio pipeline to analyze live video feeds for anomalies (e.g., density spikes, fire, structural damage) and transcribe emergency audio logs.
- **Student 2 (Real-Time Stream NLP & Scraper Agent):** Builds a high-speed ingestion engine to scrape and filter social media geotagged data, utilizing LLMs to filter noise, extract intent, and flag hyper-local emergency events.
- **Student 3 (Resource Allocation & Dispatch Agent):** Implements reasoning agents that maintain a database of real-time emergency resources (ambulances, police, barriers) and

use constraint-satisfaction algorithms alongside LLM reasoning to map out optimal deployment routes.

- **Student 4 (Central Command State & Dashboard):** Manages the overall multi-agent conversation state, updates a central GIS (Geographic Information System) interactive map dashboard, and generates automated, context-specific alerts.

Suggested Tech Stack

Vision & NLP: YOLOv10, LLaVA or Gemini Flash API

Agent Orchestration: LangChain / LangGraph

Backend & Frontend: Python, WebSockets, React.js with Mapbox/Leaflet

Database: PostgreSQL with PostGIS extension

Project Option 3: InsightAudit — Enterprise Compliance Intelligence

The Problem Statement

Large enterprises, financial institutions, and legal firms handle thousands of operational contracts, non-disclosure agreements (NDAs), and financial statements. These documents must continuously comply with rapidly changing local and international regulatory frameworks. Standard Keyword Search or simple Retrieval-Augmented Generation (RAG) systems fail because they cannot perform deep sequential reasoning, cross-reference clause dependencies across multiple documents, or provide a verifiable audit trail.

The Agentic AI Solution

An enterprise-grade platform where specialized legal/compliance agents autonomously ingest corporate documents, cross-analyze them against live-updated regulatory databases, flag non-compliance risks, and generate legally sound remediation wording.

Division of Work (4 Modules for 4 Students)

- **Student 1 (Hierarchical Document Ingestion & Advanced RAG):** Develops a chunking and indexing pipeline optimized for long-form legal documents, parsing complex tables, headers, and footnotes into a vector database with metadata filtering.
- **Student 2 (Live Regulatory Scraper & Knowledge Sync):** Builds an automated agent that monitors updates from official regulatory websites, summarizes policy updates using LLMs, and updates an internal corporate policy Knowledge Graph.

- **Student 3 (Compliance Auditor & Critic Agent):** Develops the core reasoning engine where an "Auditor Agent" flags contradictory or non-compliant clauses, and a "Critic Agent" challenges the findings to minimize false positives.
- **Student 4 (Generative Remediation Engine & UI):** Develops the component that automatically drafts revised, compliant clauses matching the tone of the original contract, and builds an enterprise dashboard displaying risk scores and side-by-side compliance audits.

Suggested Tech Stack

Frameworks: LlamaIndex and CrewAI
Vector/Graph DB: Pinecone, Milvus, or Neo4j
Document Parsing: LlamaParse or Grobid
Frontend/Backend: Python, FastAPI, React.js

Project Comparison Matrix

Project Metric	1. SentinelOps	2. ResQNet	3. InsightAudit
Primary Domain	Agentic AI & Code LLMs	Multi-Modal AI & Real-Time Data	Advanced RAG & Knowledge Graphs
Data Availability	High (Open-source repositories)	Medium (Public feeds & simulation)	High (Public templates & regulations)
System Complexity	High (Sandboxed execution)	Very High (GIS & Streams)	Medium-High (Deep reasoning)
Evaluation Impact	Clear, measurable metrics	Visual impact (Live maps)	High corporate utility